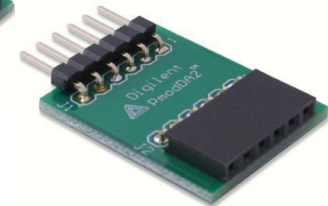
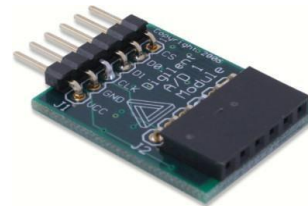
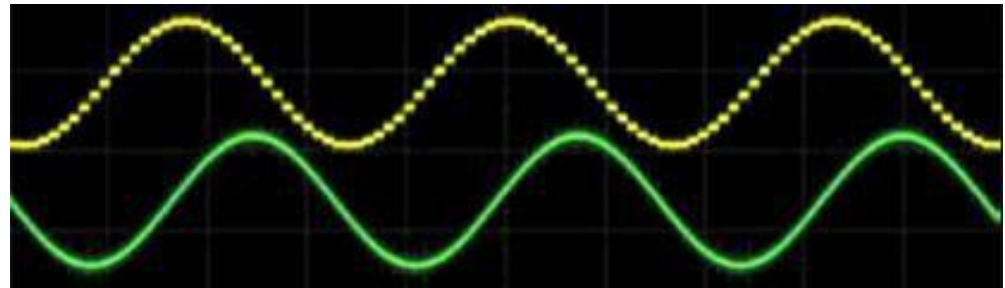


Introduction to Digital Signal Processing



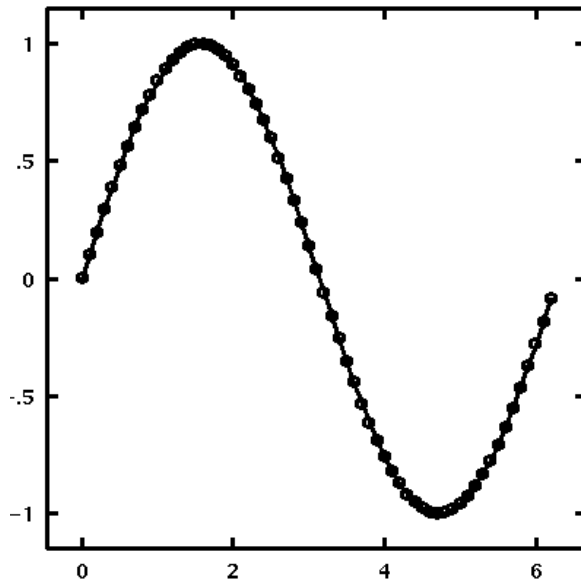
Today Agenda

- Review:
 - SPI Interface
 - ADC
 - DAC
- This Week: DPS Introduction
 - Filter
 - FIR
 - IIR

Required Reading

- *Introduction to DSP.pdf*

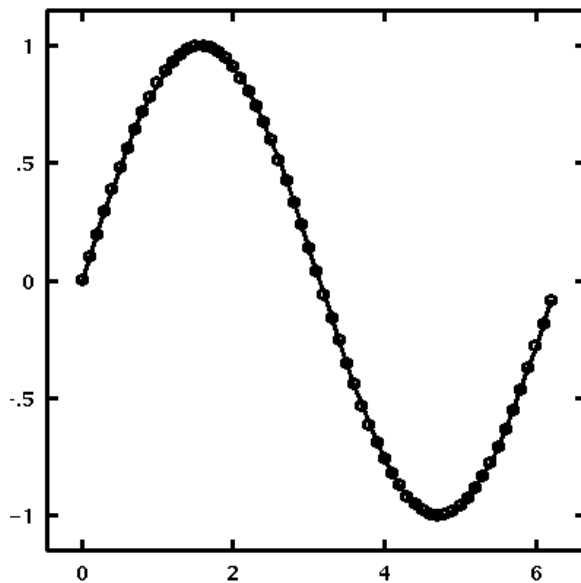
Sampling and Quantization. Analog sources of information are often derived from transducers which provide continuous electrical voltage signals from physical phenomenon such as light, pressure, temperature, vibration, and acceleration.



Position Transducer



Sampling and Quantization. These analog *baseband* signals are bandlimited to a maximum frequency. Analog baseband signals are continuous in time and amplitude and are *sampled* and *quantized* for digital signal processing (DSP).



Position Transducer



Pressure Transducer

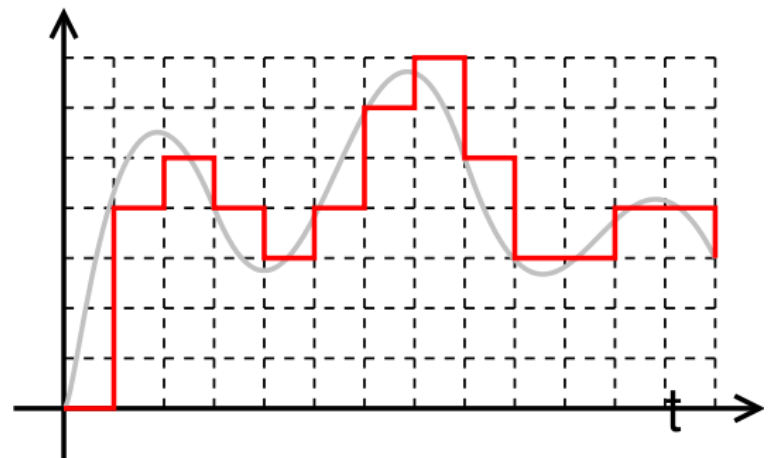


Sampling and Quantization. The output of the ideal sampler $x(mT_s)$ is inputted to an first order sample-and-hold process, whose output $y_{s-h}(t)$ is determined by:

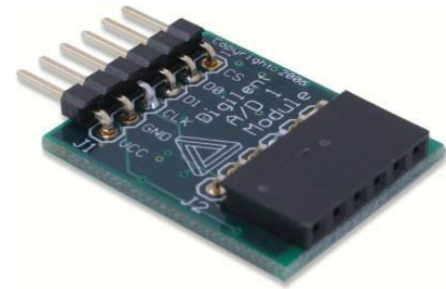
$$y_{s-h}(t) = \sum_m x(mT_s) h(t - mT_s) \quad mT_s \leq t < (m+1)T_s$$

where $h(t) = 1 \quad 0 \leq t < T_s$

$$h(t) = 0 \quad \text{otherwise}$$



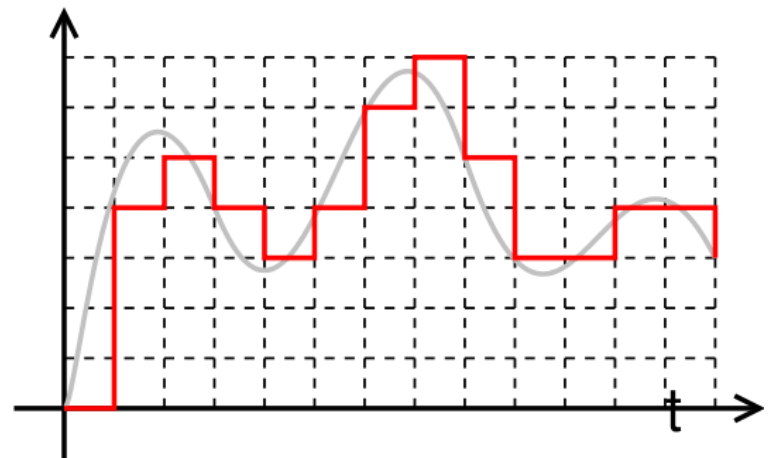
Sampling and Quantization. The sample-and-hold output $y_{s-h}(t)$ is continuous in time but represents a fixed amplitude signal during the sampling interval T_s . This analog signal is then inputted to an analog-to-digital converter (ADC) peripheral to affect the process of quantization.



$$y_{s-h}(t) = \sum_m x(mT_s) h(t - mT_s) \quad mT_s \leq t < (m+1)T_s$$

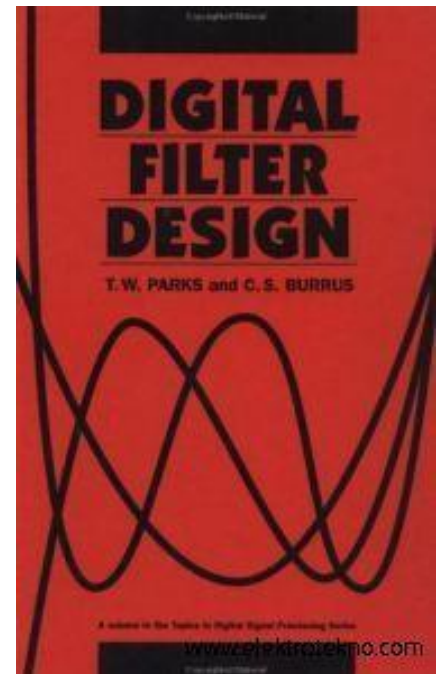
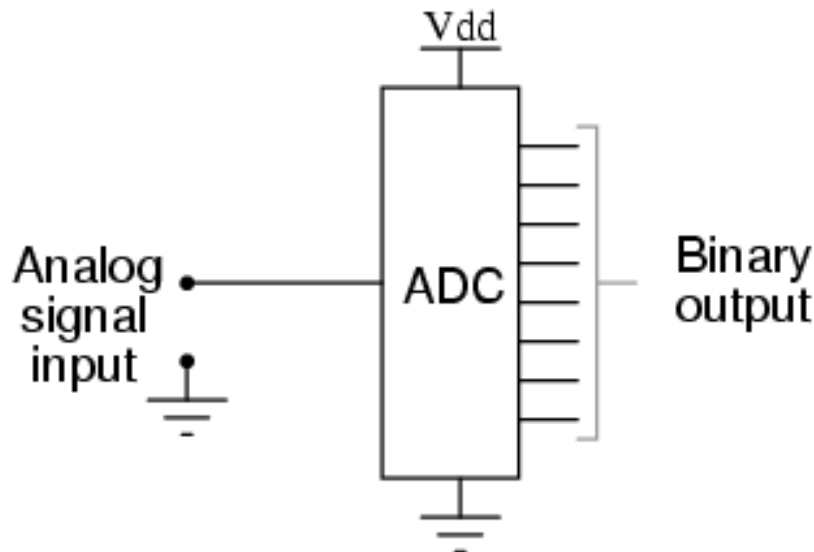
where

$$h(t) = 1 \quad 0 \leq t < T_s$$
$$h(t) = 0 \quad \text{otherwise}$$



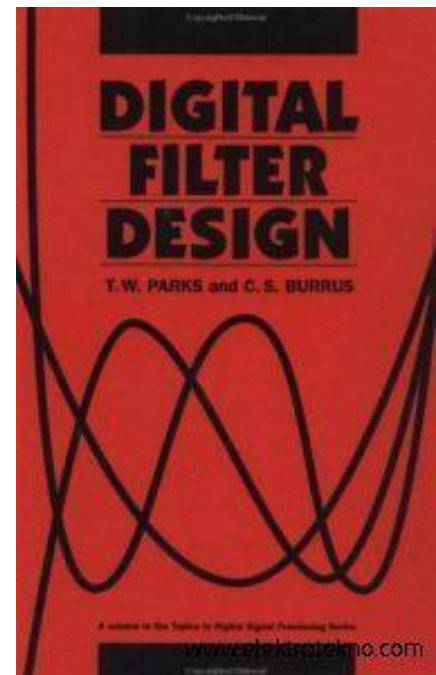
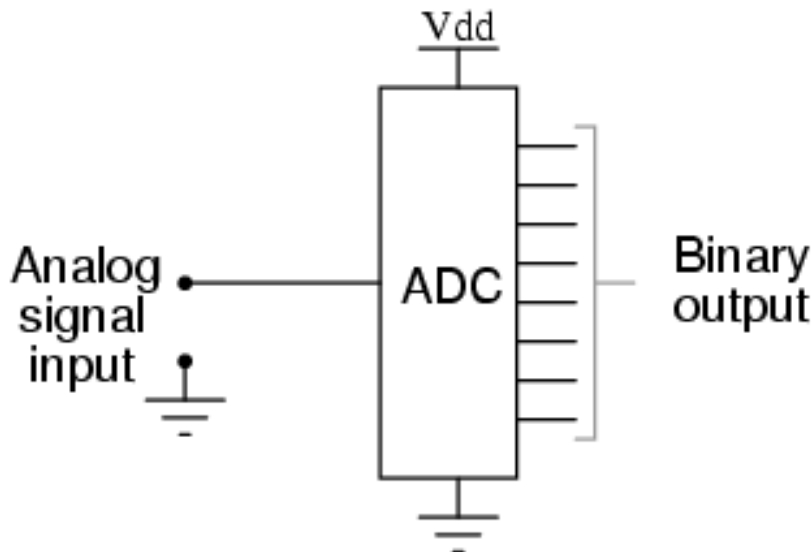
Discrete Time Sequences. The finite impulse response (FIR) class of digital filter has $a_p = 0$ for all p . FIR digital filters are also known as moving average (MA) filters since the output sequence is a weighted average of the input sequence:

$$y(m) = \sum_{q=0}^{Q-1} b_q x(m-q)$$



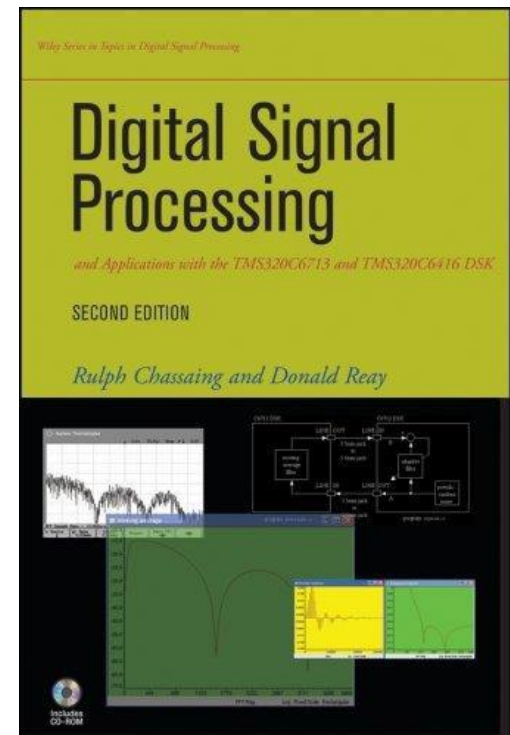
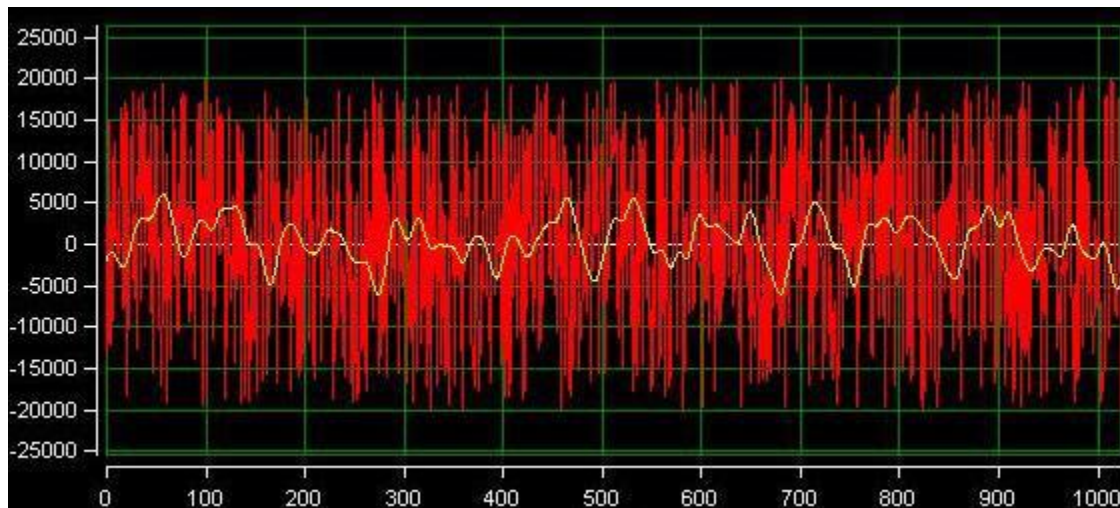
Discrete Time Sequences. The general form of the IIR digital filter is the autoregressive, moving average (ARMA) filter. Since the output sequence of the IIR digital filter at least includes a weighted sum of the past output sequences, stability of the output can be problematical.

$$y(m) = x(m) - \sum_{p=1}^{P-1} a_p y(m-p)$$



Discrete Frequency Response. DSP requires that the characteristics of the digital filter be assessed in the frequency domain. The discrete time sequence can be readily transformed to the discrete frequency domain.

$$y(m) = \sum_{q=0}^{Q-1} b_q x(m-q) + \sum_{p=0}^{P-1} a_p y(m-p)$$



Discrete Frequency Response. The discrete time input and output sequences $x(m)$ and $y(m)$ are assumed to have a discrete frequency transforms $X(z)$ and $Y(z)$, where the temporal delay d is replaced by the discrete frequency delay parameter z^{-d} , as determined by:

$$Y(z) = \sum_q^Q b_q X(z) z^{-q} + \sum_{p=1}^P a_p Y(z) z^{-p}$$

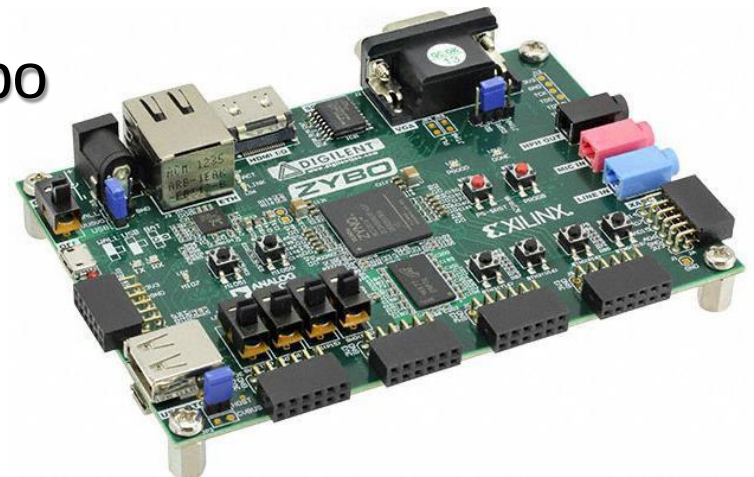
$$Y(z) - \sum_{p=1}^P a_p Y(z) z^{-p} = \sum_q^Q b_q X(z) z^{-q}$$

$$Y(z) \left[1 - \sum_{p=1}^P a_p z^{-p} \right] = X(z) \sum_q^Q b_q z^{-q}$$

Discrete Frequency Response. IIR ARMA digital filter has the discrete frequency transfer or system function $H(z) = Y(z) / X(z)$:

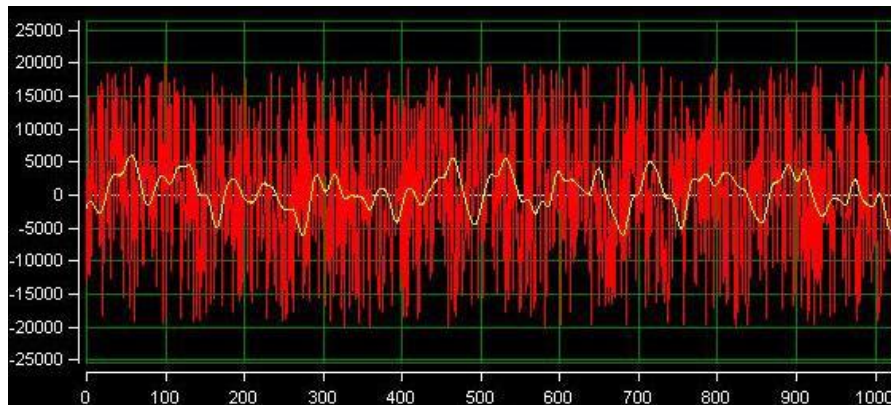
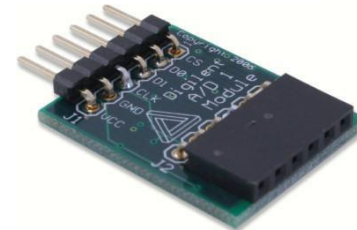
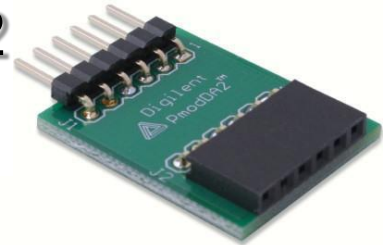
$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{q=0}^Q b_q z^{-q}}{1 - \sum_{p=1}^P a_p z^{-p}}$$

Zybo



PmodDA2

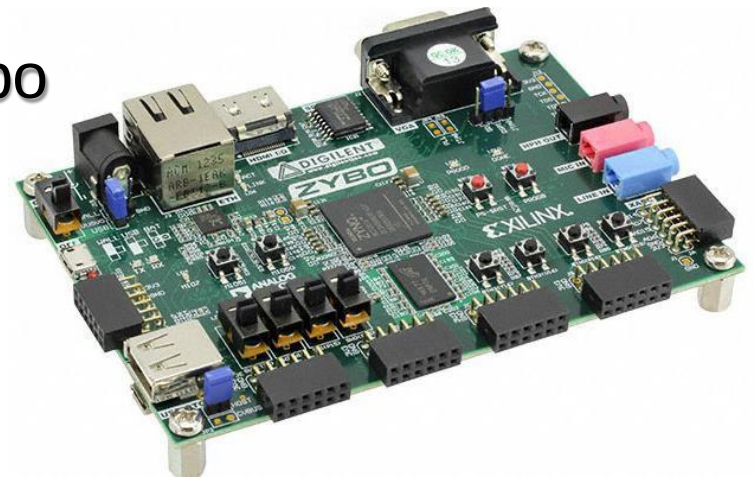
PmodAD1



Discrete Frequency Response. The discrete frequency system function $H(z)$ for the FIR or MA digital filter is then:

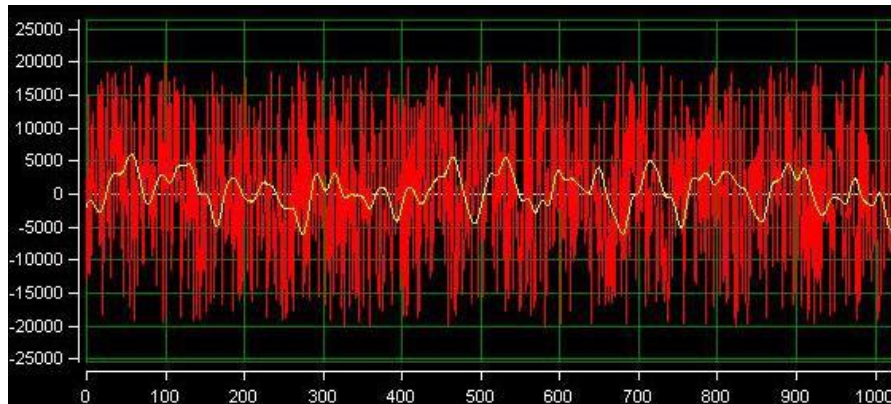
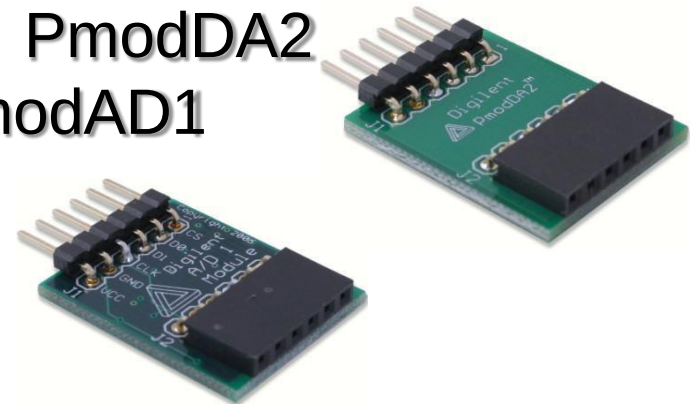
$$H(z) = \frac{Y(z)}{X(z)} = \sum_q^Q b_q z^{-q}$$

Zybo



PmodDA2

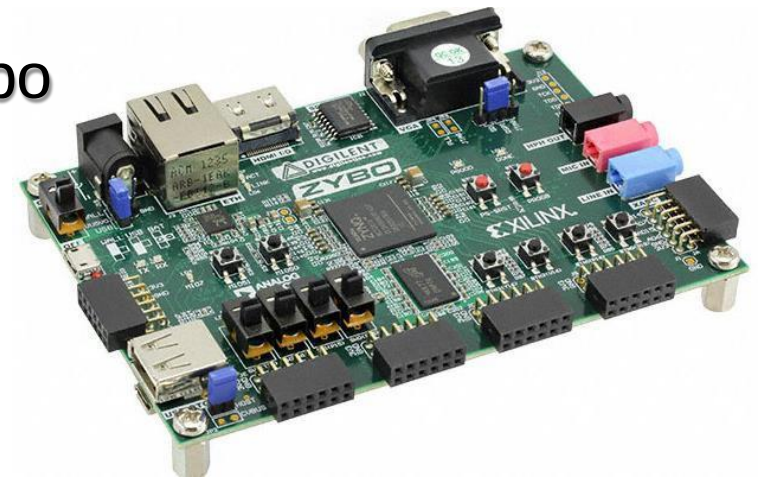
PmodAD1



Discrete Frequency Response. Finally, the discrete frequency system function $H(z)$ for the IIR AR filter is then determined by:

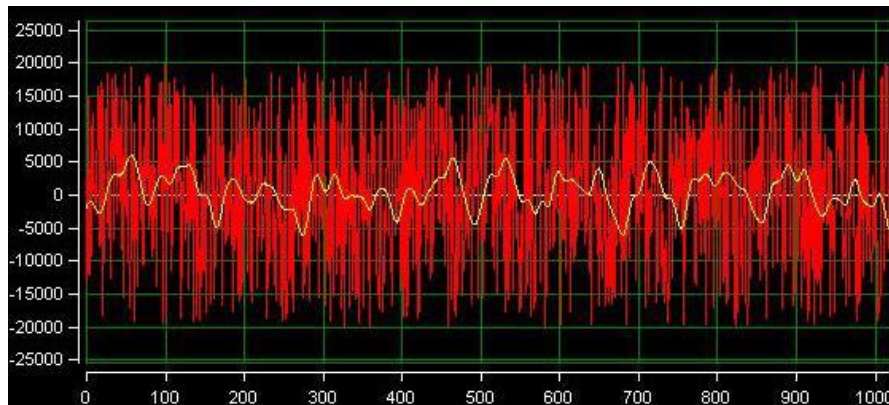
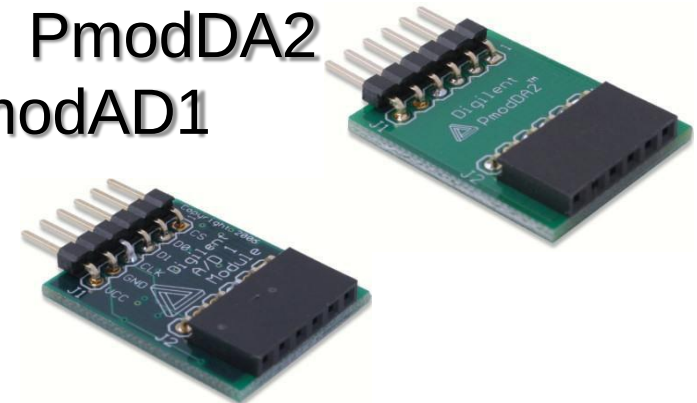
$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \sum_{p=1}^P a_p z^{-p}}$$

Zybo



PmodDA2

PmodAD1



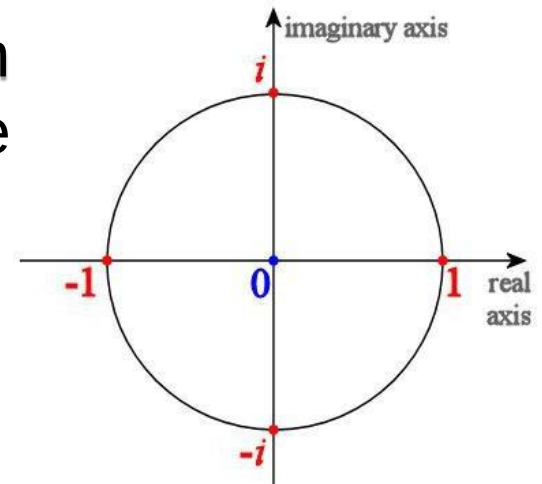
IIR Digital Filter. As a first example, a simple IIR digital filter rendered as the output sequence $y(m)$ in terms of the input sequence $x(m)$ and coefficients will illustrate the discrete frequency response analysis.

Here $Q = 0$, $q = -1$, $b_{-1} = 0.25$, $b_0 = 0.5$, $P = 1$, and $a_1 = 0.25$:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_q b_q z^{-q}}{1 - \sum_{p=1}^P a_p z^{-p}}$$

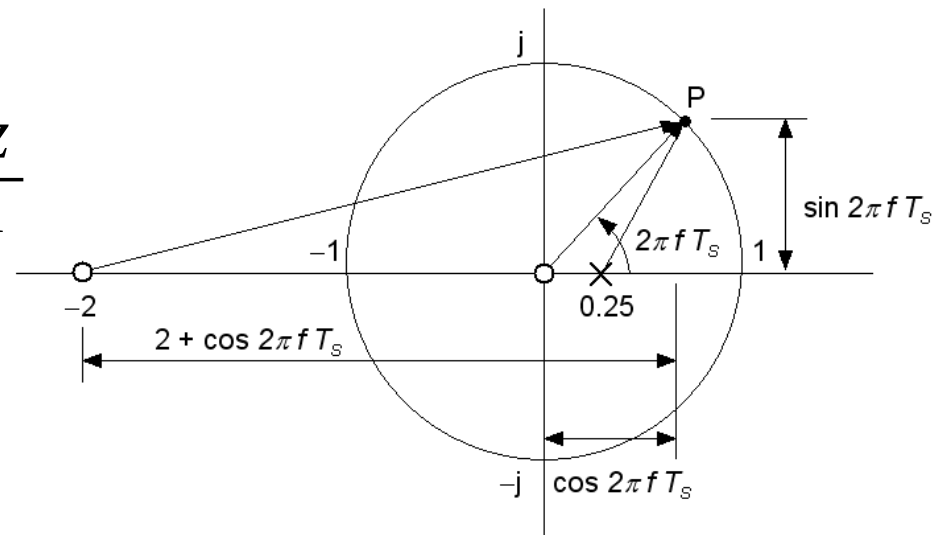
$$y(m) = 0.5 x(m) + 0.25 x(m+1) + 0.25 y(m-1)$$

IIR Digital Filter. The frequency domain for a discrete frequency transform is the *unit circle* in the complex plane. Mathematicians use i rather than j for the imaginary axis.



$$H(z) = \frac{Y(z)}{X(z)} = \frac{0.5 + 0.25 z}{1 - 0.25 z^{-1}}$$

$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$

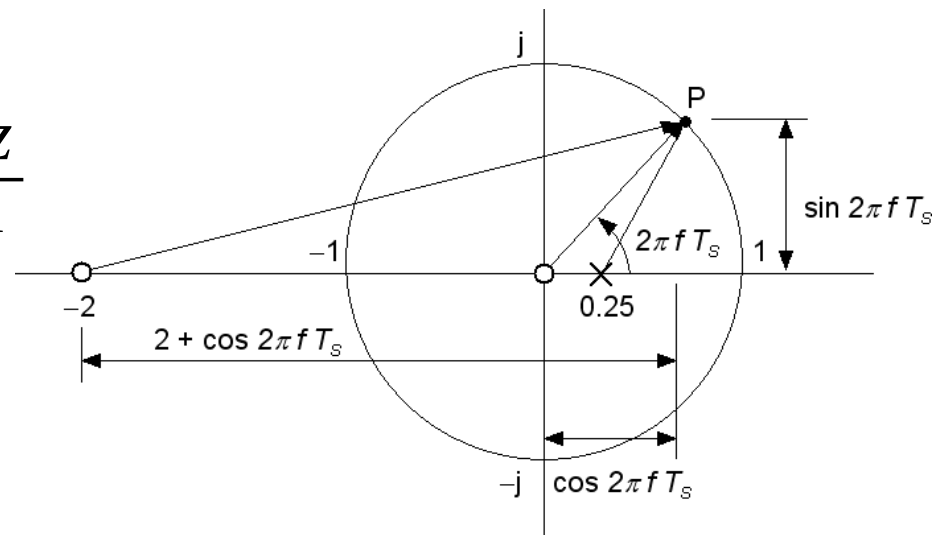


$$y(m) = 0.5 x(m) + 0.25 x(m+1) + 0.25 y(m-1)$$

IIR Digital Filter. The discrete sequence sinusoidal excitation of the discrete system $H(z)$ exists then as a point P on the unit circle in the range $0 \leq 2\pi f T_s \leq \pi$, where f is the frequency of excitation and T_s is the sampling interval.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{0.5 + 0.25 z}{1 - 0.25 z^{-1}}$$

$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$

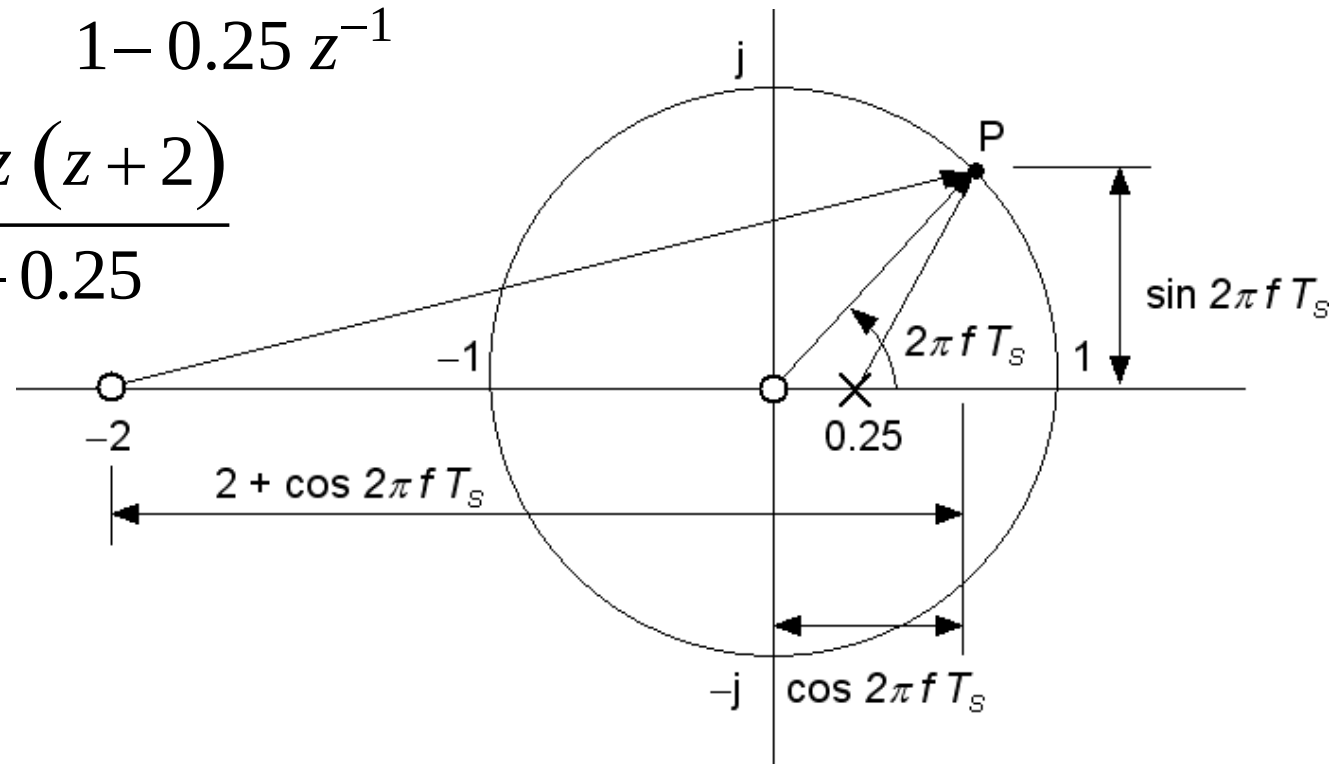


$$y(m) = 0.5 x(m) + 0.25 x(m+1) + 0.25 y(m-1)$$

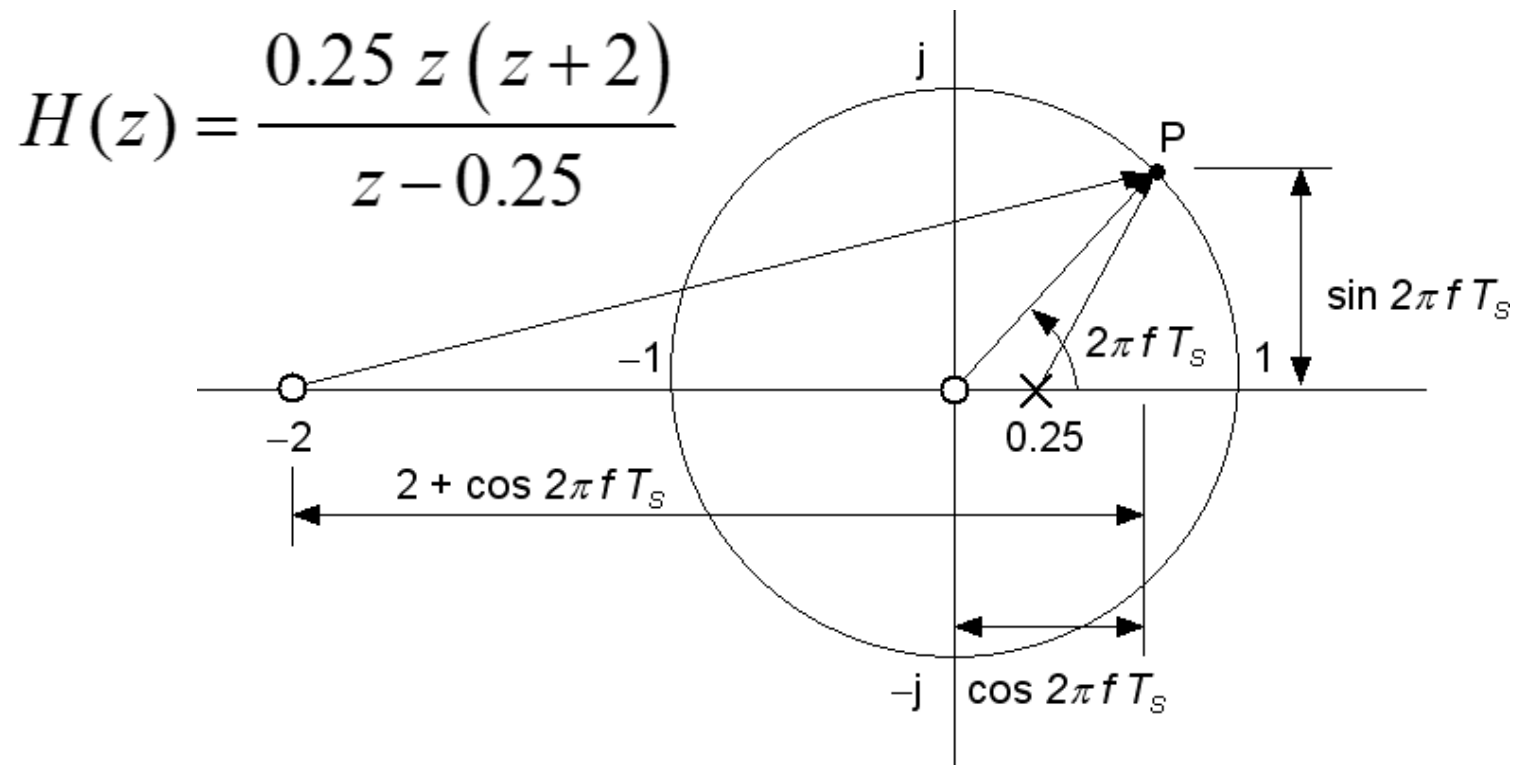
IIR Digital Filter. This simple IIR digital filter has one transfer function *pole* (where the response is infinite) at $z = 0.25$, two transfer function *zeros* (where the response is zero) at $z = 0$ and $z = -2$ and a multiplicative term of 0.25.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{0.5 + 0.25 z}{1 - 0.25 z^{-1}}$$

$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$

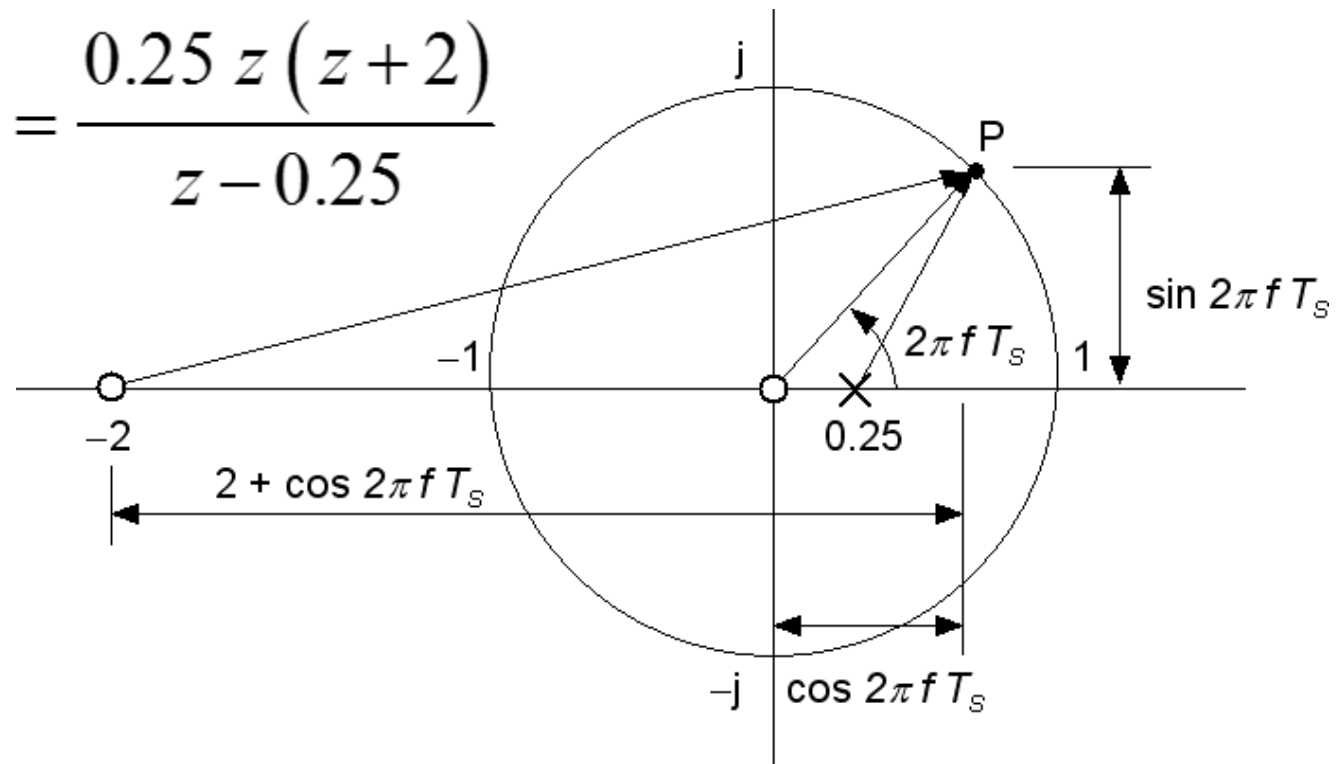


IIR Digital Filter. The pole (indicated by $\times$) and the zeros (indicated by $\circ$) of the transfer function are plotted on the complex frequency domain plane as shown.

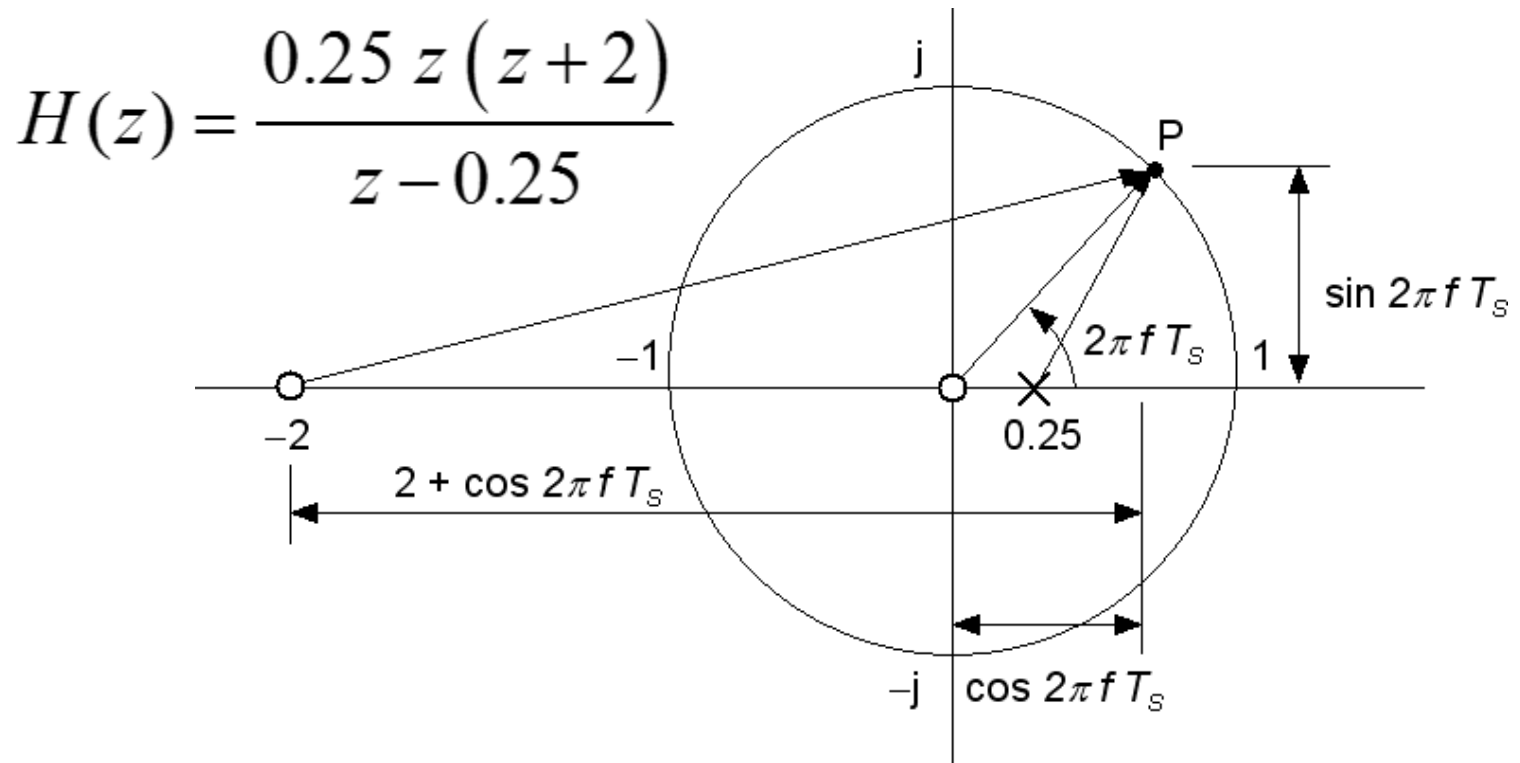


IIR Digital Filter. Vectors are drawn from the pole and zeros to an arbitrary point P on the unit circle which represents the sinusoidal excitation of the discrete system $H(z)$.

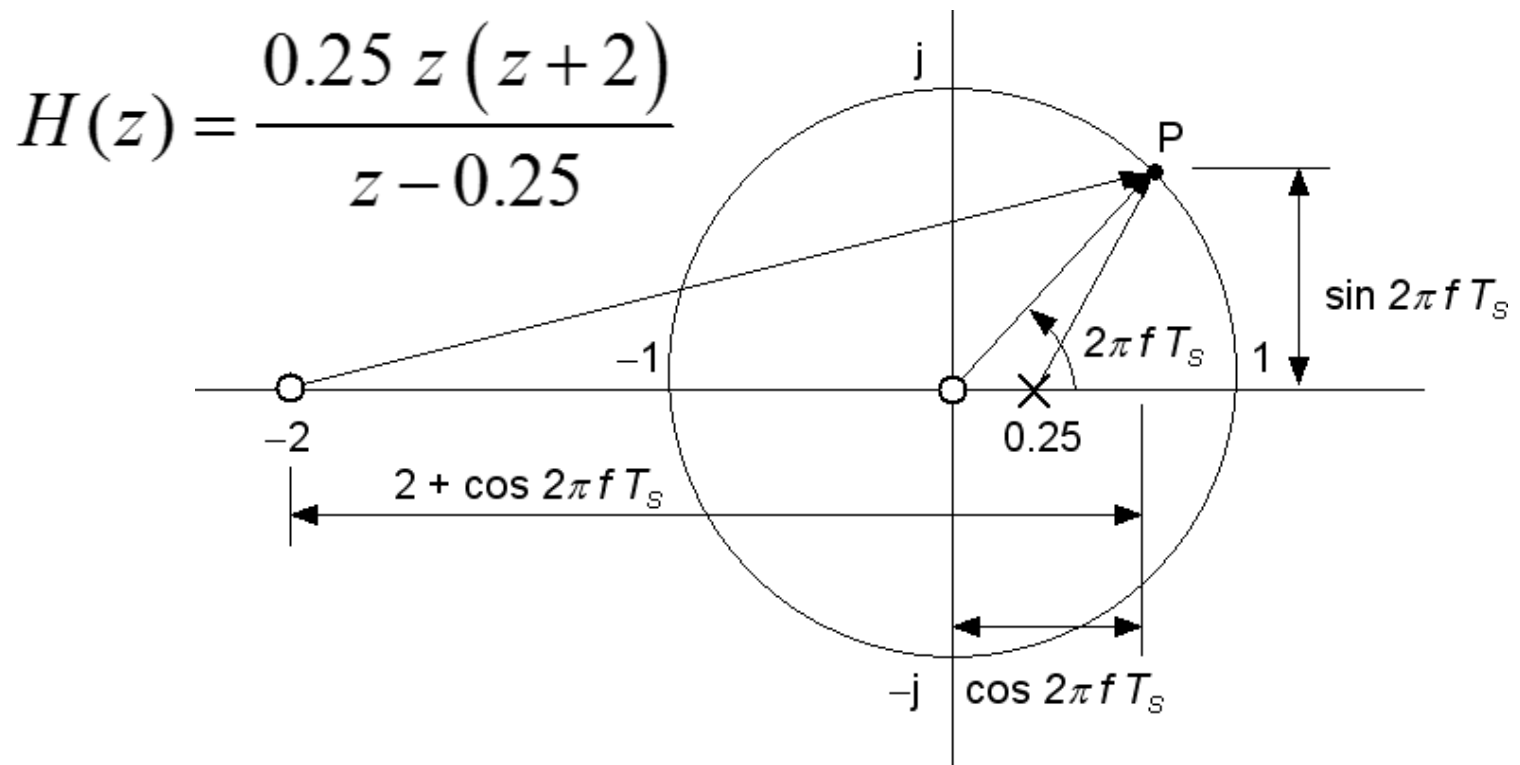
$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$



IIR Digital Filter. The magnitude of the discrete frequency response $|H(z)|$ is determined by the product of the gain G and the length of the vector(s) from the zero(s) divided by the product of the length of the vector(s) from the poles.



IIR Digital Filter. The angle from the origin of the complex plane to the point P on the unit circle where the sinusoidal excitation occurs is in the range $0 \leq 2\pi f T_s \leq \pi$.



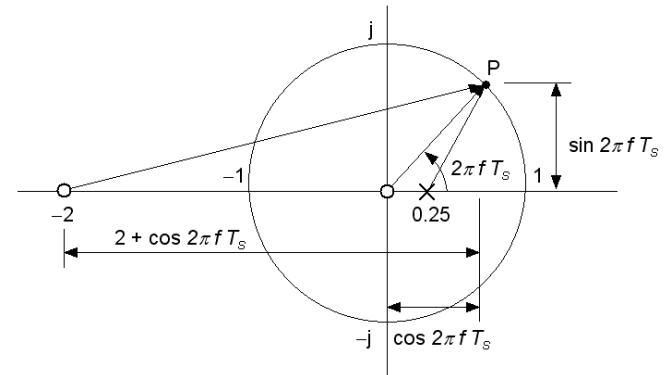
IIR Digital Filter. The simple geometry allows the length of these vectors and the magnitude of the discrete frequency response to be determined:

$$G = 0.25 \quad L_{z=0} = 1$$

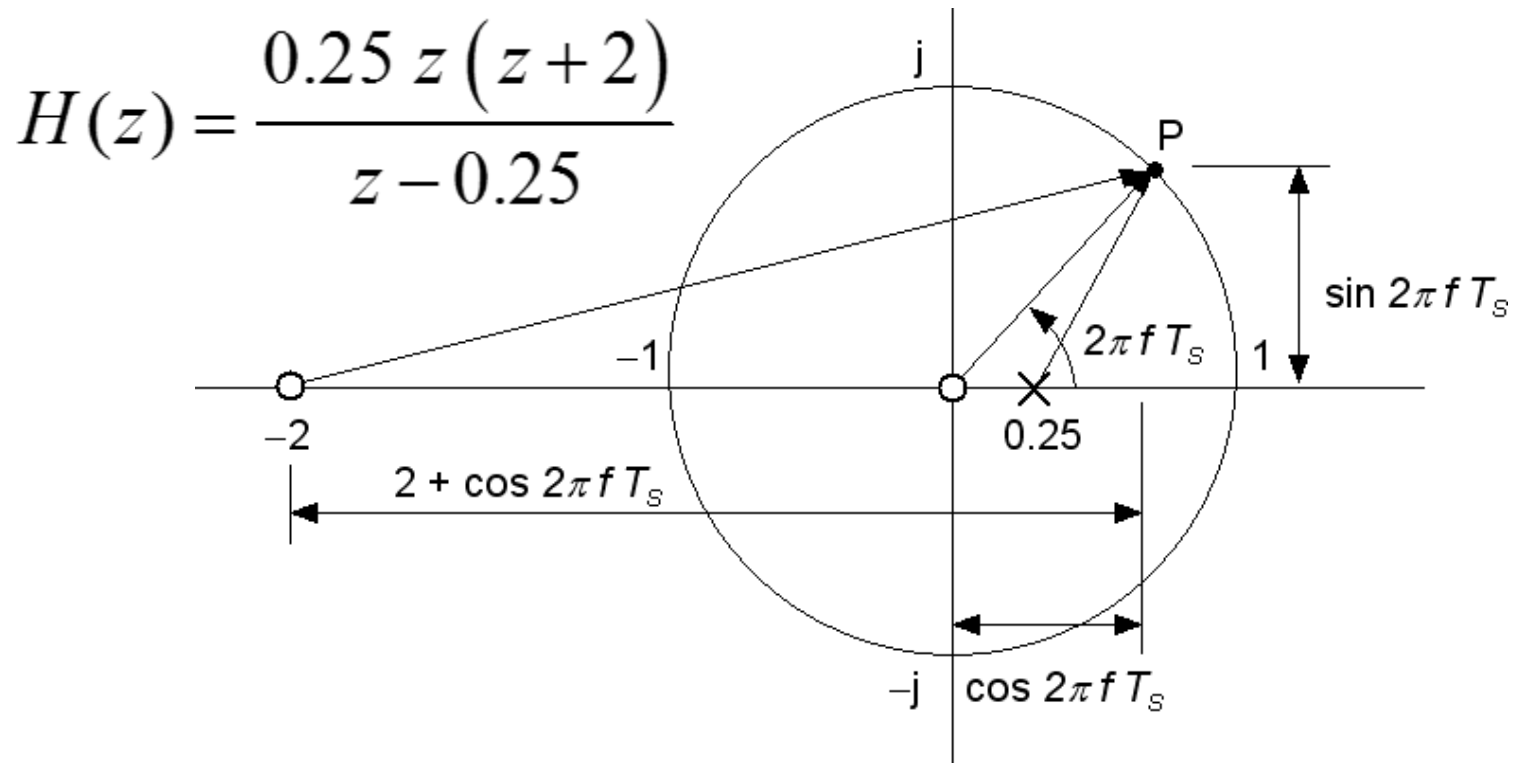
$$L_{z=-2} = \sqrt{(2 + \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s}$$

$$L_{p=0.25} = \sqrt{(\cos 2\pi f T_s - 0.25)^2 + \sin^2 2\pi f T_s}$$

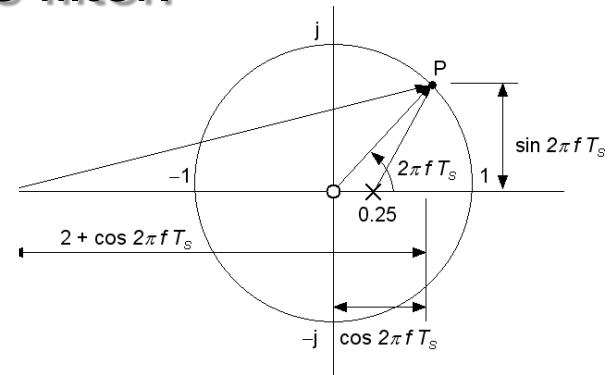
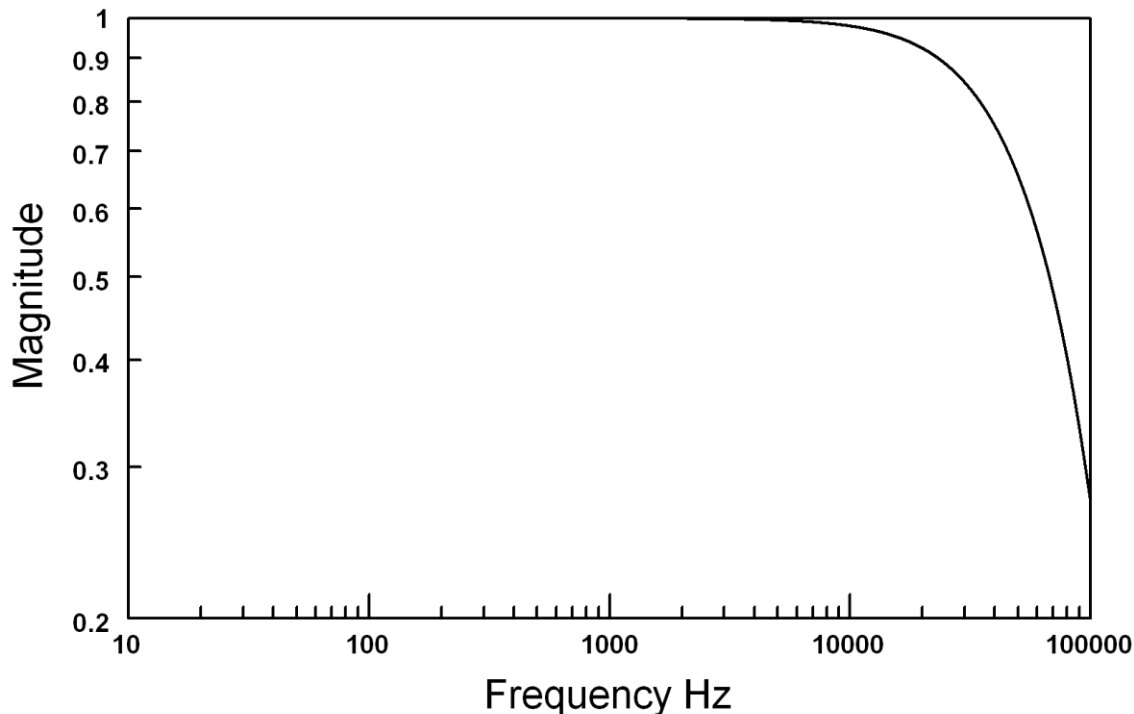
$$|H(z)| = 0.25 \frac{\sqrt{(2 + \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s}}{\sqrt{(\cos 2\pi f T_s - 0.25)^2 + \sin^2 2\pi f T_s}}$$



IIR Digital Filter. This IIR digital filter has a *low pass* characteristic with a -3 dB (magnitude = 0.707) cutoff frequency of approximately 44.5 kHz and a low frequency gain of 1.



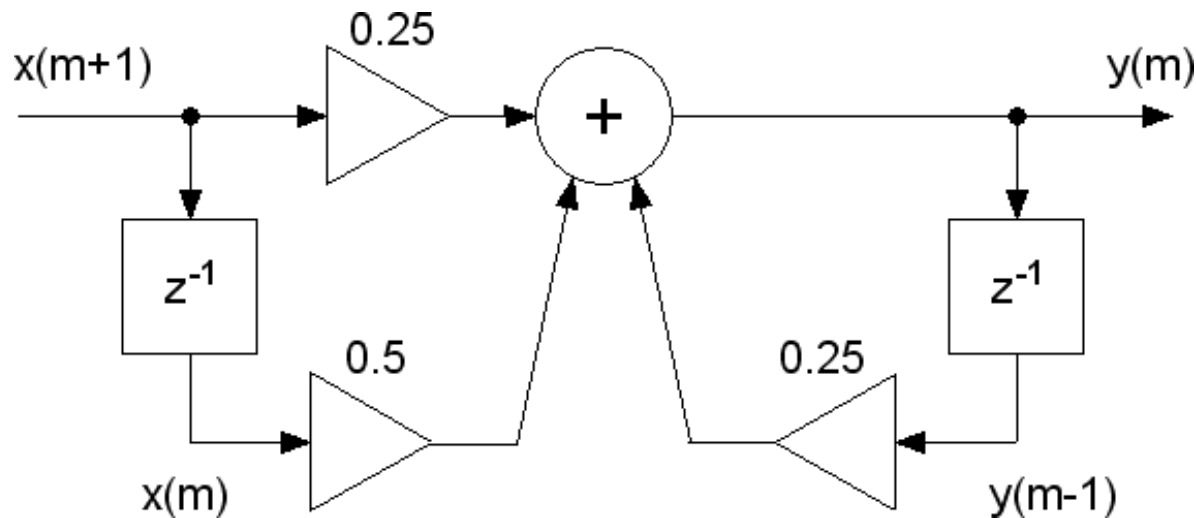
IIR Digital Filter. The magnitude of the discrete frequency response of this IIR digital filter is shown on a *log-log* scale plotted against frequency Hz with $T_s = 4 \times 10^{-6}$ sec or $f_s = 1/T_s = 250$ kHz and is a low pass filter.



$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$

IIR Digital Filter. The IIR low pass digital filter from can be rendered as a *building block* in a *direct* form as shown.

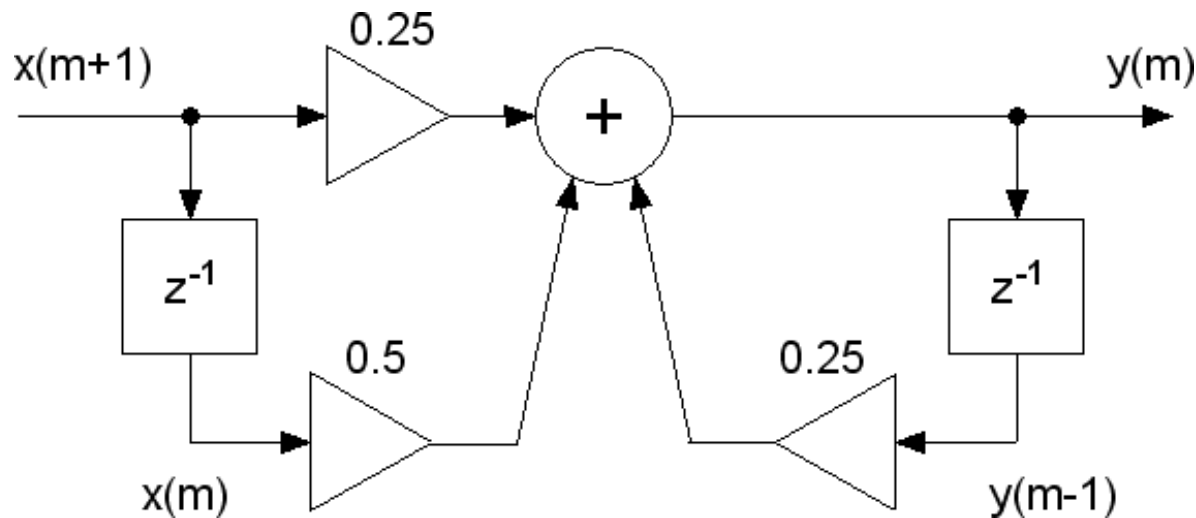
$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$



$$y(m) = 0.5 x(m) + 0.25 x(m+1) + 0.25 y(m-1)$$

IIR Digital Filter. The building block uses delay elements (z^{-1}) from storage registers, the gain elements from multipliers and an adder.

$$H(z) = \frac{0.25 z (z + 2)}{z - 0.25}$$



$$y(m) = 0.5 x(m) + 0.25 x(m+1) + 0.25 y(m-1)$$

FIR Digital Filter. As a second example, a simple FIR digital filter rendered as the output sequence $y(m)$ in terms of the input sequence $x(m)$ and coefficients will illustrate the discrete frequency response analysis.

Here $Q = 1$, $q = -1$, $b_{-1} = -0.5$, $b_0 = 2$ and $b_1 = -1.5$

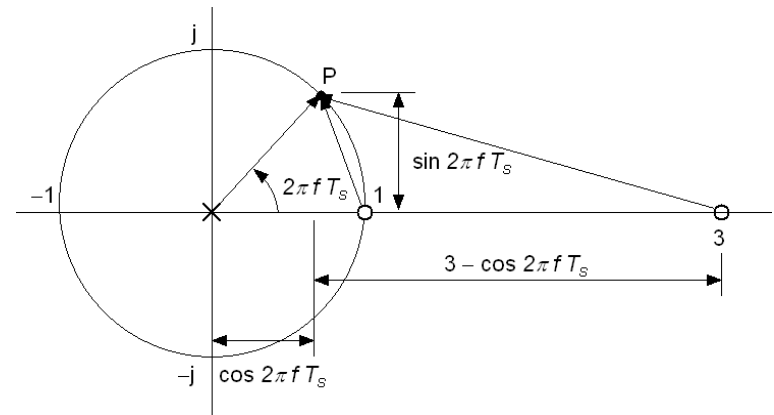
$$H(z) = \frac{Y(z)}{X(z)} = \sum_q^Q b_q z^{-q}$$

$$y(m) = 2 x(m) - 1.5 x(m-1) - 0.5 x(m+1)$$

FIR Digital Filter. The discrete frequency transfer function $H(z)$ is:

$$H(z) = \frac{Y(z)}{X(z)} = 2 - 1.5 z^{-1} - 0.5 z = \frac{2z - 1.5 - 0.5z^2}{z}$$

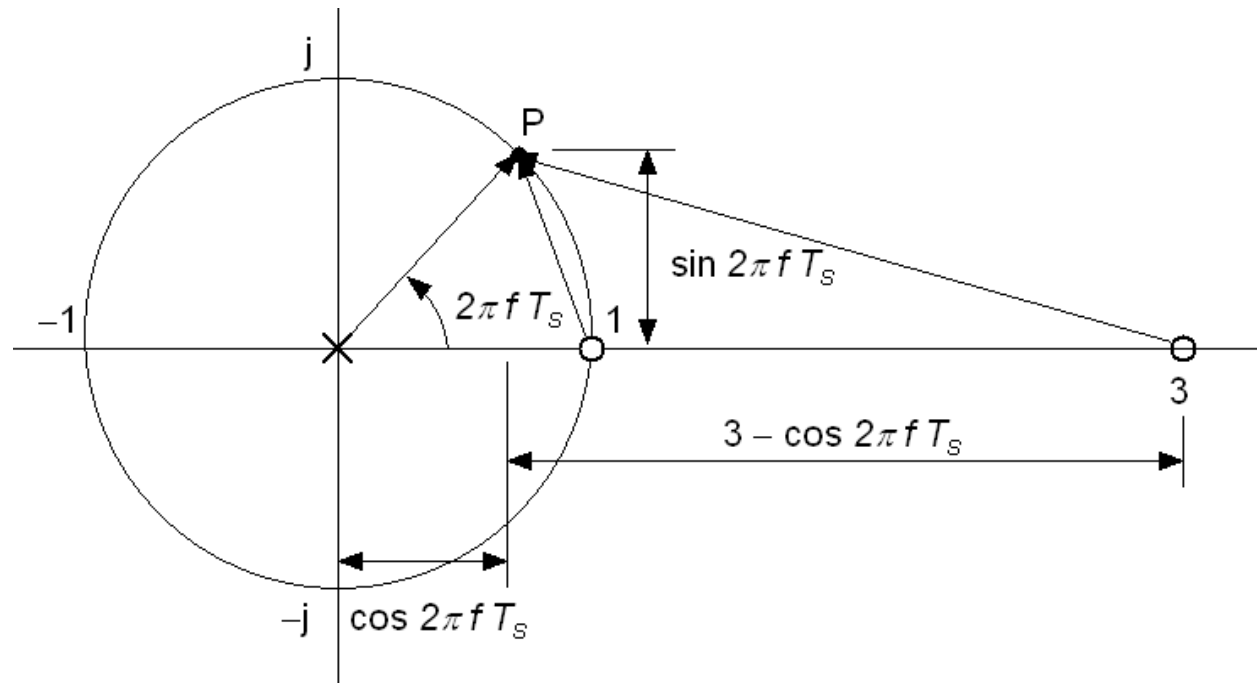
$$H(z) = \frac{0.5(z - 3)(1 - z)}{z}$$



$$y(m) = 2 x(m) - 1.5 x(m - 1) - 0.5 x(m + 1)$$

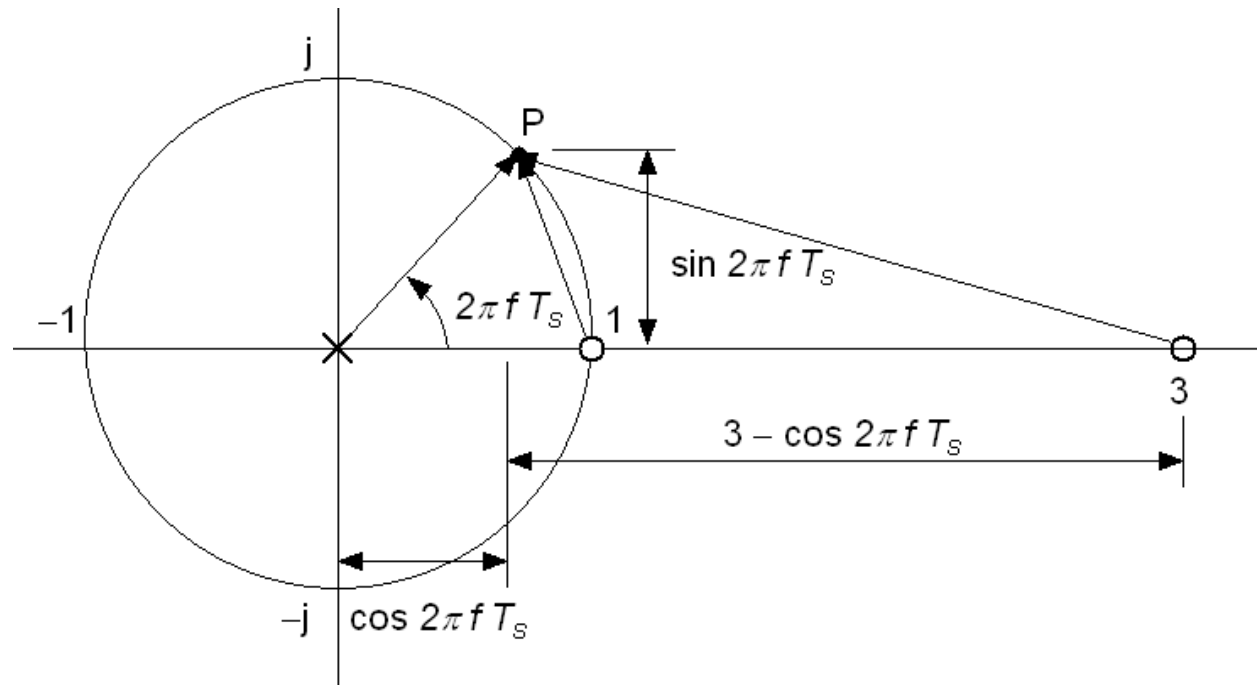
FIR Digital Filter. This simple FIR digital filter has one transfer function *pole* (where the response is infinite) at $z = 0$, two transfer function zeros (where the response is zero) at $z = 3$ and $z = 1$ and a multiplicative term of 0.5.

$$H(z) = \frac{0.5 (z - 3) (1 - z)}{z}$$



FIR Digital Filter. The pole (indicated by $\times$) and the zeros (indicated by $\circ$) of the transfer function are plotted on the complex frequency domain plane as shown.

$$H(z) = \frac{0.5 (z - 3) (1 - z)}{z}$$



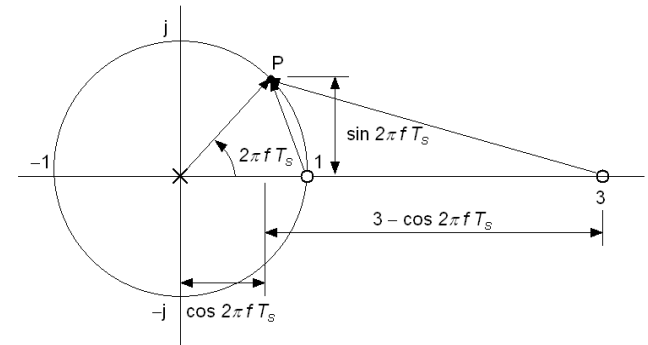
FIR Digital Filter. As for the IIR digital filter, the magnitude of the discrete frequency response $|H(z)|$ is determined by the product of the gain G and the length of the vector(s) from the zero(s) divided by the product of the length of the vector(s) from the poles.

$$G = 0.5 \quad L_{p=0} = 1$$

$$L_{z=3} = \sqrt{(3 - \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s}$$

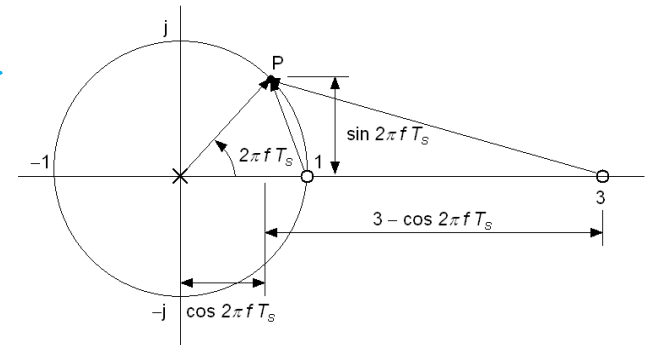
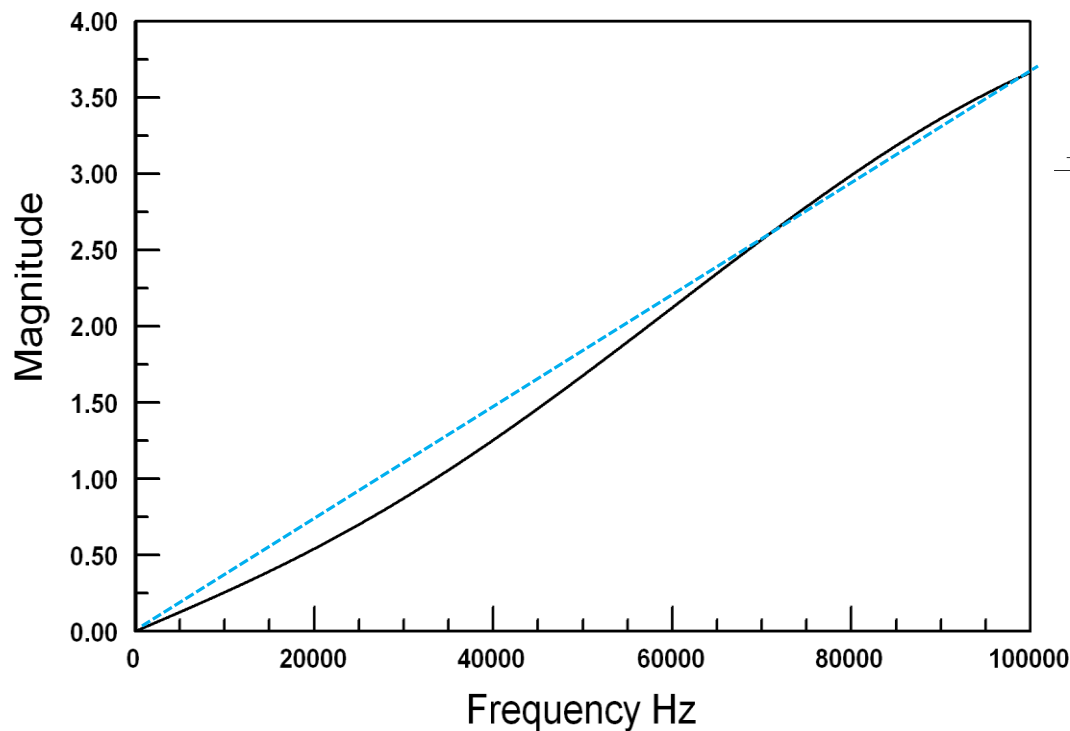
$$L_{z=1} = \sqrt{(1 - \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s}$$

$$|H(z)| = 0.5 \sqrt{(3 - \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s} \times \sqrt{(1 - \cos 2\pi f T_s)^2 + \sin^2 2\pi f T_s}$$



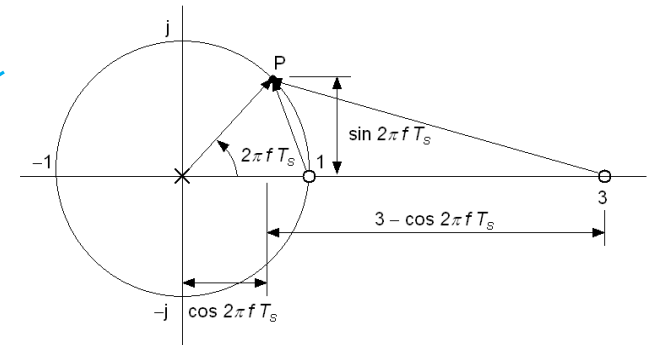
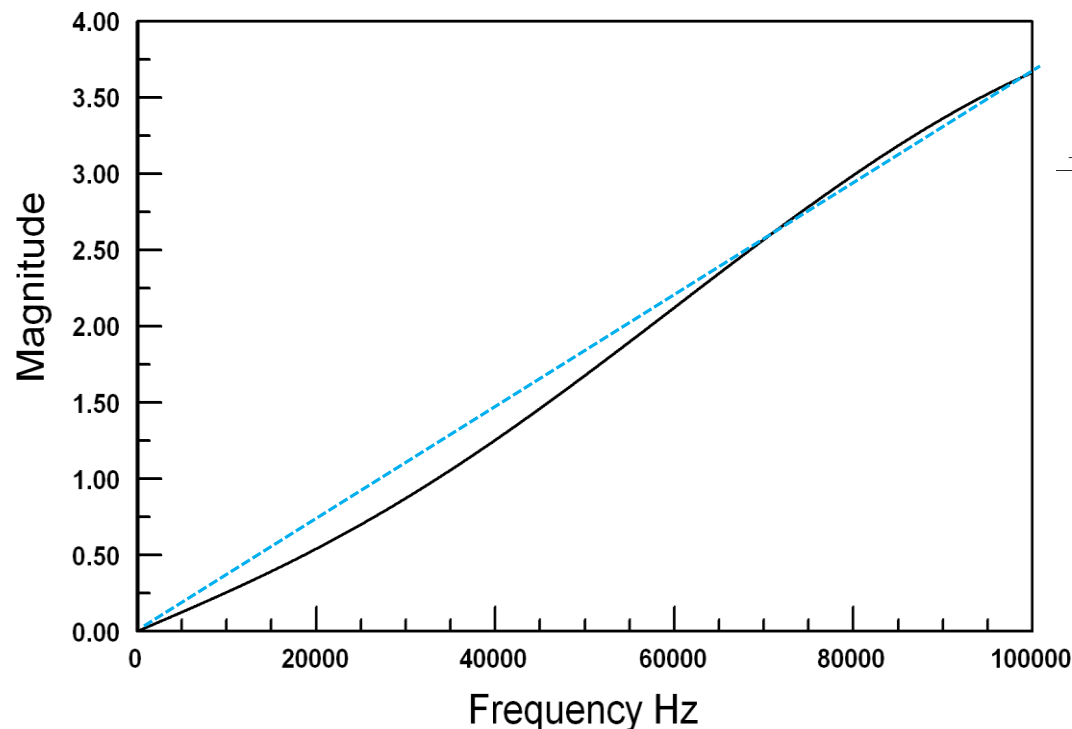
FIR Digital Filter. The magnitude of the discrete frequency response of this FIR digital filter is shown on a *log-log* scale plotted against frequency Hz with $T_s = 4 \times 10^{-6}$ sec or $f_s = 1/T_s = 250$ kHz.

$$H(z) = \frac{0.5 (z - 3) (1 - z)}{z}$$



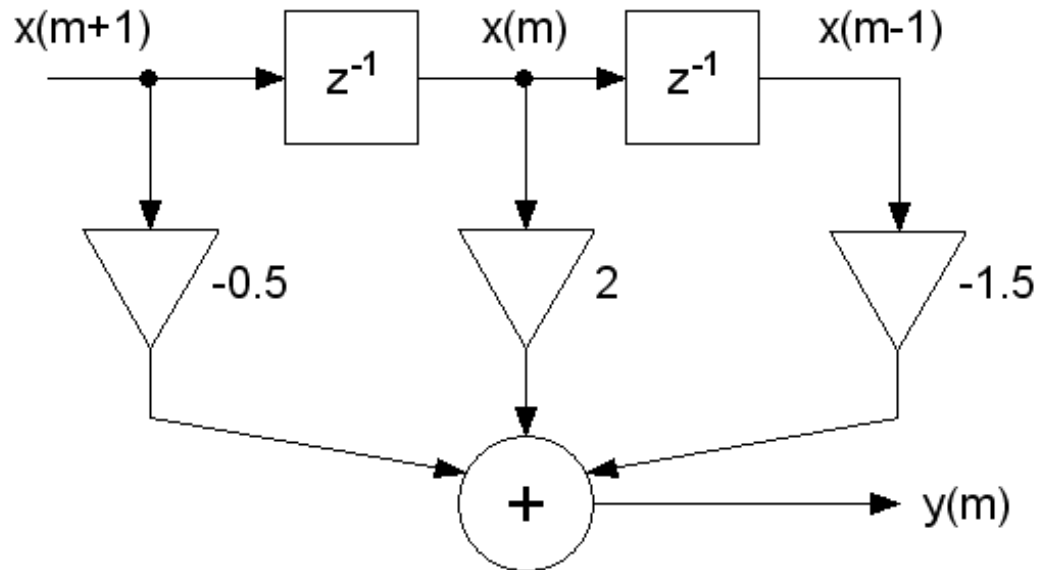
FIR Digital Filter. This FIR digital filter has a *derivating* (d/dt) characteristic because of its approximately linear magnitude response with frequency.

$$H(z) = \frac{0.5 (z - 3) (1 - z)}{z}$$



FIR Digital Filter. The FIR derivating digital filter can be rendered as a building block in a direct or *transversal* form as shown.

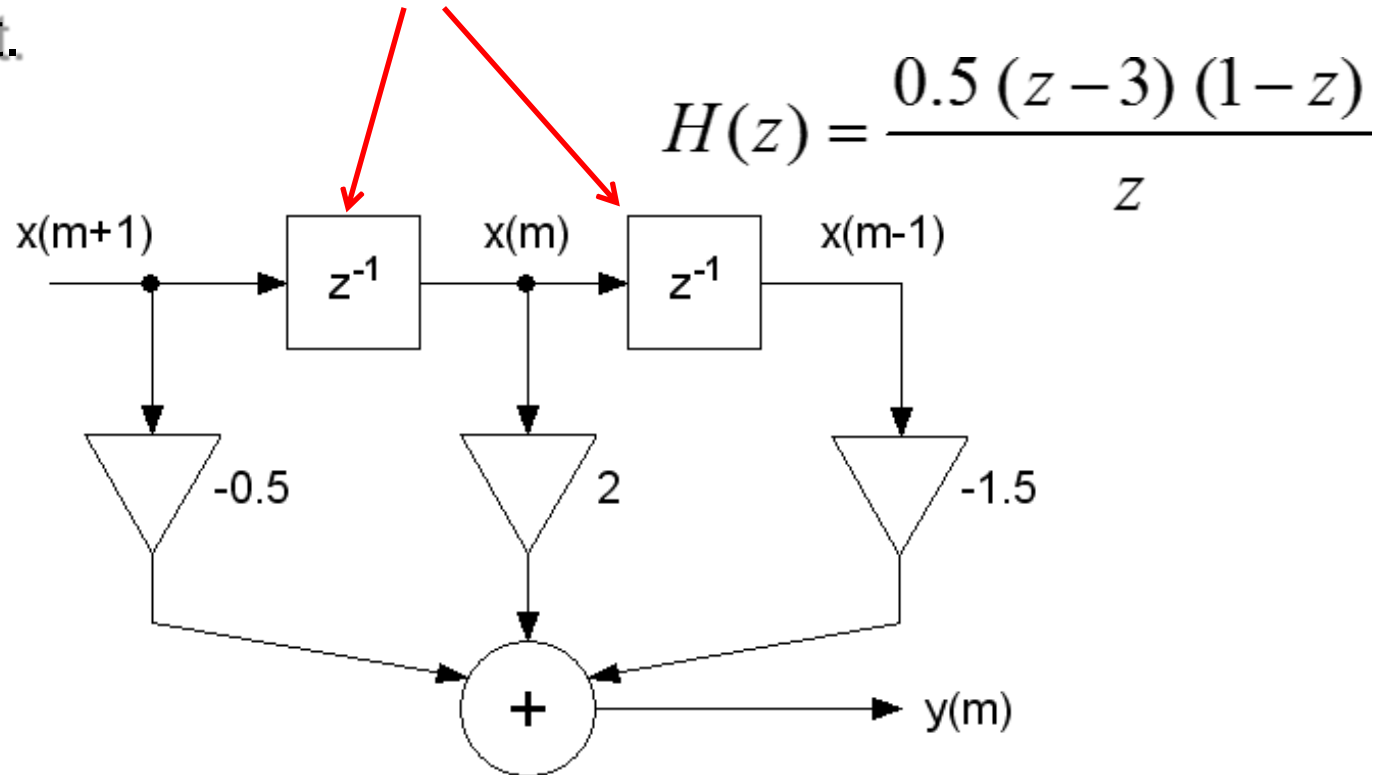
$$H(z) = \frac{0.5 (z - 3) (1 - z)}{z}$$



Eq. 4.20

$$y(m) = 2 x(m) - 1.5 x(m - 1) - 0.5 x(m + 1)$$

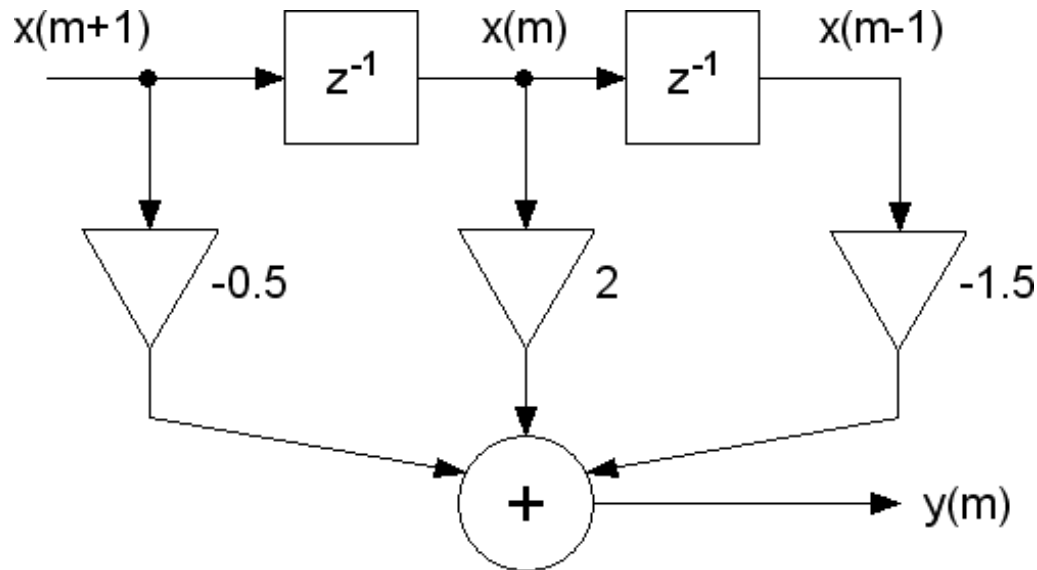
FIR Digital Filter. The transversal FIR digital filter represents a *tapped delay line* and is particularly simple to implement.



Eq. 4.20

$$y(m) = 2 x(m) - 1.5 x(m-1) - 0.5 x(m+1)$$

FIR Digital Filter. The building block uses delay elements (z^{-1}) from storage registers, the gain elements from multipliers and an adder/subtractor.



$$y(m) = 2 x(m) - 1.5 x(m-1) - 0.5 x(m+1)$$

Digital Filters

You have received an audio file with a secret message. The message cannot be deciphered because of an interference signal centered at 10KHz.

1. Assuming that the received signal is already digitized, and the sampling frequency is 40KHz, design an appropriate IIR BANDSTOP digital filter having (2 poles each with magnitude (0.7), and 2 zeros each with magnitude (0.7)) to attenuate the interference signal. Specifically:
 - a. [10 points] What would be the exact locations of your poles and zeros on the Z-plane?
 - b. Calculate the Transfer Function.
 - c. Calculate the Difference Equation.

End of Introduction to Digital Signal Processing

